25900283

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCCHE200 - INORGANIC CHEMISTRY - I
(2024 ADMISSION ONWARDS)

Instruction to the Examiners:

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.

Part A
Multiple choice questions, Answer all questions Each question carries 1 mark.

1. **Which of the following is a correct match?**
Ans: c). sp – linear
2. **The shared pairs of electrons are called**……
Ans: a). Bond Pairs
3. **Based on its MO diagram, what is the bond order and magnetic behavior of the N_2 molecule?**
Ans: a). Bond order 3, diamagnetic
4. **Fajan's rule is applicable mainly to**
Ans**:** b). Ionic compounds with polarizable ions
5. **Pitchblende is** ----------
Ans**:** a) UO_2
6. **Which actinide ion in +3 oxidation state has just half-filled 5f-subshell?**
Ans**:** Ans**:** c) Cm^{3+}
7**. The carbon -14 dating method is based on the fact that**
Ans**:** a) Ratio of carbon-14 and carbon-12 is constant
8. **The oxidation state in Co in $[Co(NH_3)_6]Cl_3$ is**
Ans**:** b). +3
9. **A chelating ligand must have:**
Ans**:** b). Two or more donor atoms
10. **According to Werner's theory, the primary valency of a metal in a coordination compound is always satisfied by:**
Ans**:** b). Negative ions

Part B
Short Answer Questions, Answer any 4 questions, Each question carries 3 marks

11. Limitations of Valence Bond Theory (VBT)
Ans: 1.VBT considers that electron pairs forming a covalent bond are localized between two atoms. In reality, some molecules (e.g., benzene) have delocalized π-electrons, which VBT cannot explain, 2.Failure to Explain Magnetic Properties, 3.No Explanation for Color of Complexes,4. Inability to Explain Spectral Properties of coordination compounds, 5. Fails to Explain the Stability and Reactivity of metal complexes,6. No Consideration of Molecular Orbitals (Any 3 Limitations -Give 3 marks)

12. Define hybridization and state two of its key characteristics.
Ans: Hybridization is the process of mixing atomic orbitals of nearly equal energy on the same atom to form new equivalent orbitals called hybrid orbitals. These hybrid orbitals are used to form covalent bonds in molecules.
Key Characteristics of Hybridization
1. The number of hybrid orbitals formed is equal to the number of atomic orbitals mixed.
1. 2. Hybrid orbitals are identical in shape and energy, and they orient in space to minimize repulsion, resulting in specific molecular geometries. (Definition- 1 mark, two characteristics-2 marks)

13. Why is lattice energy not measured directly and how does the Born Haber cycle help to determine it?

Ans: Lattice energy is the energy released when one mole of an ionic compound in crystalline form is formed from the constituent ions in the gaseous phase. However some of the ionic compounds especially alkali metal halides are formed from the reaction of an alkali metal with a gaseous halogen. Therefore the lattice energy is not a quantity that can be measured directly. In the Born–Haber cycle, the formation of an ionic solid is considered as taking place via the direct formation of the substance from the metal and the non-metal, and the other through a series of changes from the same constituents. By Hess's law, the energy change should be the same for both processes. This means that the overall energy change in a process depends only on the energy of the initial and final states, and not on the routes taken. (For correct explanation – Give 3 marks)

14. A molecule of KBr has a bond length of 2.82 A° and an observed dipole moment of 10.41D. Calculate the percentage ionic character of the bond

Bond length (r) = 2.82 Å = 2.82×10^{-10} m , Observed dipole moment = 10.41 D

Percentage ionic character = $(\mu_{obs} / \mu_{theoretical}) \times 100$

$\mu_{theoretical} = e \times r = (1.6 \times 10^{-19}\ C) \times (2.82 \times 10^{-10}\ m) = 4.512 \times 10^{-29}\ C{\cdot}m$

Convert to Debye: 1 D = 3.336×10^{-30} C·m, $\mu_{theoretical} = (4.512 \times 10^{-29}) / (3.336 \times 10^{-30}) = 13.53$ D

Percentage ionic character = $(\mu_{obs} / \mu_{theoretical}) \times 100$

$= (10.41 / 13.53) \times 100 = 76.94\%$

(Only equation – 1 mark; calculation till $\mu_{theoretical}$– 1.5 marks; Whole calculation till %ionic character and answer- Give 3 marks)

15. Discuss the effect of lanthanide contraction on separation of lanthanides.

Ans: The steady decrease in atomic size of lanthanoids with increase in atomic number is called Lanthanide contraction. Poor shielding effect of 4f- electrons- consequently, lanthanoids have only a small change in ionic radii making their separation in pure state difficult.

(Justification – 2 marks. Statement of difficulty in separation of lanthanides – 1 mark)

16. Calculate CFSE for a Tetrahedral complex having d7 electronic configuration

Ans: The electronic configuration is $e^4 t_2^3$.

CFSE = $4 \times (-0.6\ \Delta_t) + 3 \times (+0.4\ \Delta_t)$

$= (-2.4 + 1.2)\ \Delta_t = -1.2\ \Delta_t$ (Only equation – 1 mark; Correct calculation with answer-Give 3 marks)

Part C

Short Essay Questions, Answer any 2 questions, Each question carries 7 marks

17. Derive the Born-Landé equation used for calculating the lattice energy of ionic compounds

Ans: The Born–Landé equation is a means of calculating the lattice energy of a crystalline ionic compound. Lattice energy of an ionic crystal is determined by Coulombic interactions and they are of two types:

Attractive interactions between ions of opposite charges, and Repulsive interactions due to interpenetration of similarly charged electron charge clouds. The electrostatic energy between two oppositely charged ions is given by:

$$E = \frac{z^+ z^- e^2}{r}$$

where z^+ and z^- are the charges on positive and negative ions respectively, e is the charge on an ion

For 1 mole,

$$E_A = -\frac{N_A M z^+ z^- e^2}{r}$$

Where N_A is the Avogadro number and M is the Madelung constant, which depends upon the geometry of the crystal. Born and Landé suggested that potential energy due to repulsion is inversely proportional to the nth power of the distance between the ions; the repulsion energy between the ions would be proportional

to $1/r^n$ (where r is the distance between the ions), so that the repulsive energy term, E_R, would be expressed as:

$$E_R = \frac{N_A B}{r^n} \qquad \text{where, B = constant}$$

The total energy holding the crystal together is the lattice energy (U).

$$U = E_A + E_R = -\frac{N_A M z^+ z^- e^2}{r} + \frac{N_A B}{r^n}$$

The attractive and repulsive energies are balanced at the equilibrium distance where the crystal lattice will be stable. At equilibrium, the potential energy of an ion will be minimum, i.e., dU/dr = 0, and the equilibrium distance = r_0. Differentiating with respect to r, will lead to

$$B = \frac{M z^+ z^- e^2 r_0^{n-1}}{n}$$

Putting this value of B in the equation for U at the state of equilibrium:

$$U = -\frac{N_A M z^+ z^- e^2}{r_0} + \frac{N_A}{r_0^n}\frac{M z^+ z^- e^2 r_0^{n-1}}{n}$$

$$U = -\frac{N_A M z^+ z^- e^2}{r_0}\left(1 - \frac{1}{n}\right)$$

This equation is known as the Born–Landé equation. With this equation, the lattice energy can be calculated from knowledge of the geometry of the crystal, the charges of the cation and anion, and the interionic distance. When expressed in SI units, the equation takes the form:

$$E = -\frac{N_A M z^+ z^- e^2}{4\pi\epsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

Where ε_0 = permittivity of free space. Since n is always greater than 1, U will always be negative as expected.
(Correct derivation and final answer-Give 7 marks; Only → Born-Landé equation-1 mark, equation to determine lattice energy – 1 mark)

18. Discuss the role of chain reactions in nuclear energy. How are they controlled in nuclear reactors?

Ans: Relevant explanation for chain reaction with example – 3 marks,
distinction between controlled and uncontrolled chain reaction(4 marks).
(Since second part of the question is beyond the scope of the syllabus give entire 7 marks to first part of the question.)

19. Compare the stability of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$. Explain your answer.

Ans: In both complexes the oxidation state of iron is +3 (Fe^{3+}) and the electronic configuration of $Fe^{3+}$is d^5. The CN^- is a strong-field ligand which produces a large octahedral splitting and it tends to give low-spin complexes. The F^- is a weak-field ligand which produces a small splitting and it tends to give high-spin complexes. For low-spin d^5 $[Fe(CN)_6]^{3-}$, all five electrons occupy the t_{2g} orbitals, CFSE = $5 \times (-0.4\Delta_0) = -2.0\ \Delta_0$. For high-spin d^5 $[FeF_6]^{3-}$, the configuration is $t_{2g}^3\ eg^2$, CFSE = $3\times(-0.4\Delta_0) + 2\times(+0.6\Delta_0) = 0.0\ \Delta_0$. Therefore, $[Fe(CN)_6]^{3-}$ gains an additional stabilization of ≈ $2.0\ \Delta_0$ compared to $[FeF_6]^{3-}$ due to ligand field effects. (Correct answer with explanation - Give 7 marks)

Part D

Essay Type Questions; Answer any 1 questions, Each question carries 14 marks

20. Define half-life and decay constant. Derive the mathematical relationship between them and discuss their significance in the study of radioactivity

Ans: Decay constant – Definition – **2 marks**,
$\lambda = \left(\frac{2.303}{t}\right) log\left[\frac{a}{a-x}\right]or\lambda = \left(\frac{2.303}{t}\right) log\left[\frac{N_0}{N_t}\right]$-Equation **: (1 mark),**
Definitions for half life period – **2 marks**
$t_{\left(\frac{1}{2}\right)} = \left(\frac{0.693}{\lambda}\right)$ - Equation : **(1 mark**),
Derivation:
Decay constant :Upto$-\left(\frac{dx}{a-x}\right) = \lambda dt$ **(2 marks)**
Up to $\ln(a-x) = -\lambda t + c$ **(1mark)**
Upto$\lambda = \left(\frac{2.303}{t}\right) log\left[\frac{a}{a-x}\right]$ **(2marks)**

Half life period: Derivation :$\lambda = \left(\frac{2.303}{t_{\left(\frac{1}{2}\right)}}\right) log\left[\frac{a}{a-\left(\frac{a}{2}\right)}\right]$**(1 mark)**

Upto$\lambda = \left(\frac{2.303}{t_{\left(\frac{1}{2}\right)}}\right) log[2]$ **(1mark)**

Upto$t_{\left(\frac{1}{2}\right)} = \left(\frac{0.693}{\lambda}\right)$**(1 mark)**

21. Explain the general characteristics of d block elements.

1. Variable Oxidation States, 2. Formation of Coloured Compounds, 3. Paramagnetic Behaviour, 4. Formation of Complex Ions 5. Catalytic Properties, 6. Formation of Interstitial Compounds, 7. Alloy Formation, 8. High Melting and Boiling Points, 9. Metallic Character, 10. Formation of Coloured Ions and Complexes
(Any 5 characteristics with explanation-Give 14 marks)
